

L-R Separation from Chiral Localization

Connecting d_{LR} to 5D Domain-Wall Fermions

EDC Project

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Abstract

We analyze the chiral localization of fermions in a 5D brane-world setup to understand the origin of the L-R center separation d_{LR} . For a single symmetric domain wall, both chiralities localize at the wall center, giving $d_{\text{LR}} = 0$. Non-zero separation requires additional structure: either two walls, asymmetric profiles, or junction geometry. The OPR-21 gate constraint $I_4 \sim 10^{-3}$ GeV forces $d_{\text{LR}}/\delta \sim 5\text{--}10$. The coincidence $d_{\text{LR}} \approx r_p$ (proton radius) suggests junction geometry as the underlying mechanism, but this remains a hypothesis [\[P\]](#), not a derivation.

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1 The 5D Dirac Equation

1.1 Setup

Definition 1.1 (5D Dirac Action). [\[Der\]](#) The 5D Dirac action for a fermion Ψ coupled to a domain-wall mass is:

$$S = \int d^5x \sqrt{-g} [\bar{\Psi} (i\Gamma^M D_M - m(\chi)) \Psi] \quad (1)$$

where χ is the extra-dimensional coordinate, Γ^M are 5D gamma matrices, and $m(\chi)$ is a position-dependent mass (domain-wall profile).

1.2 Chiral Decomposition

In 5D, there is no intrinsic chirality. However, for a 4D observer, we can decompose the 5D spinor into left- and right-handed 4D components:

$$\Psi(x^\mu, \chi) = \psi_L(x^\mu) w_L(\chi) + \psi_R(x^\mu) w_R(\chi) \quad (2)$$

where $\gamma_5 \psi_{L,R} = \mp \psi_{L,R}$ (4D chirality).

Proposition 1.2 (Mode Equations). [\[Der\]](#) The transverse profiles satisfy coupled first-order equations:

$$\partial_\chi w_L + m(\chi) w_L = \lambda w_R \quad (3)$$

$$-\partial_\chi w_R + m(\chi) w_R = \lambda w_L \quad (4)$$

For massless 4D modes ($\lambda = 0$):

$$\partial_\chi w_L^{(0)} = -m(\chi) w_L^{(0)} \quad (5)$$

$$\partial_\chi w_R^{(0)} = +m(\chi) w_R^{(0)} \quad (6)$$

Proof. Standard dimensional reduction of the 5D Dirac equation. $\square$

2 Single Domain Wall: $d_{LR} = 0$

2.1 The Tanh Profile

Definition 2.1 (Domain-Wall Mass). [\[Dc\]](#) A standard domain-wall profile is:

$$m(\chi) = \mu \tanh\left(\frac{\chi}{\delta}\right) \quad (7)$$

where μ is the bulk mass scale and δ is the wall thickness.

2.2 Zero Mode Solutions

Theorem 2.2 (Zero Mode Profiles). [\[Der\]](#) For the domain-wall mass (7), the normalizable zero modes are:

$$w_L^{(0)}(\chi) \propto \exp\left(-\int_0^\chi m(\chi') d\chi'\right) = \left[\cosh\left(\frac{\chi}{\delta}\right)\right]^{-\mu\delta} \quad (8)$$

$$w_R^{(0)}(\chi) \propto \exp\left(+\int_0^\chi m(\chi') d\chi'\right) = \left[\cosh\left(\frac{\chi}{\delta}\right)\right]^{+\mu\delta} \quad (9)$$

Proof. Direct integration of (5) and (6) using $\int \tanh(x) dx = \ln \cosh(x)$. $\square$

Corollary 2.3 (Normalizability). *[Der] For $\mu > 0$:*

- $w_L^{(0)}$ is normalizable: $\int |w_L^{(0)}|^2 d\chi < \infty$
- $w_R^{(0)}$ is NOT normalizable: diverges as $|\chi| \rightarrow \infty$

For $\mu < 0$: roles are reversed.

Key result: A single domain wall supports ONE chiral zero mode, not both.

2.3 Peak Position Analysis

Proposition 2.4 (Peak at Wall Center). *[Der] For the normalizable mode $w_L^{(0)}$ (with $\mu > 0$):*

$$\left. \frac{d|w_L^{(0)}|^2}{d\chi} \right|_{\chi=0} = 0 \quad (10)$$

The mode peaks at $\chi = 0$ (wall center).

Proof. $|w_L^{(0)}|^2 \propto [\cosh(\chi/\delta)]^{-2\mu\delta}$ which is maximized at $\chi = 0$ where $\cosh(0) = 1$ is minimal. $\square$

Critical Result: Single Wall Gives $d_{LR} = 0$

For a single symmetric domain wall:

- Only ONE chirality has a normalizable zero mode
- That mode peaks at the wall center $\chi = 0$
- The opposite chirality has no localized mode (or peaks elsewhere)

Therefore: A single domain wall CANNOT produce separated L and R zero modes. To get $d_{LR} > 0$, we need additional structure.

3 Mechanisms for $d_{LR} > 0$

To achieve L-R separation, several mechanisms are possible:

3.1 Mechanism A: Two Domain Walls

Definition 3.1 (Two-Wall Configuration). *[Dc] Place two domain walls at $\chi = \pm d/2$:*

$$m(\chi) = \mu \left[\tanh\left(\frac{\chi - d/2}{\delta}\right) - \tanh\left(\frac{\chi + d/2}{\delta}\right) - 1 \right] \quad (11)$$

In this configuration:

- w_L localizes on wall at $\chi = +d/2$
- w_R localizes on wall at $\chi = -d/2$ (or vice versa)

- Separation: $d_{\text{LR}} = d$

Status: This is the “brane-localized fermions” approach of Randall-Sundrum and similar models. The separation d is a free parameter of the geometry — it must be determined by other physics.

3.2 Mechanism B: Yukawa Coupling to Higgs Profile

Definition 3.2 (Yukawa Mechanism). **[Dc]** If there is a Higgs-like field $\phi(\chi)$ with Yukawa coupling:

$$\mathcal{L}_{\text{Yuk}} = -y\phi(\chi)\bar{\psi}_L\psi_R + \text{h.c.} \quad (12)$$

and $\phi(\chi)$ has a non-trivial profile, this modifies the effective localization positions.

If $\phi(\chi) \propto \chi$ (linear VEV gradient), the L and R modes experience different effective potentials and localize at different positions.

Status: This requires specifying the Higgs profile $\phi(\chi)$, which introduces additional parameters.

3.3 Mechanism C: EDC Junction Geometry

Postulate 3.3 (Y-Junction Localization). **[P]** In the EDC framework, the proton is a Y-junction with three arms meeting at 120° . The characteristic size is the proton charge radius:

$$r_p \approx 0.84 \text{ fm} \quad (13)$$

Proposition 3.4 (Junction-Induced Separation). **[P]** If L-modes localize at the junction center and R-modes localize at the arm tips (or vice versa), the separation is:

$$d_{\text{LR}} \sim r_p \approx 0.84 \text{ fm} \quad (14)$$

Status: This is a hypothesis **[P]**, not a derivation. It explains the $d_{\text{LR}} \approx r_p$ coincidence but requires showing that junction geometry actually produces this localization pattern.

4 Constraint from OPR-21 Gates

4.1 The I_4 Gate

Definition 4.1 (Overlap Integral). **[Der]** The four-point overlap controlling G_F is:

$$I_4 = \int d\chi w_L^2(\chi) w_R^2(\chi) w_\phi^2(\chi) \quad (15)$$

where w_ϕ is the mediator mode profile.

Proposition 4.2 (Exponential Sensitivity). **[Der]** For Gaussian-like modes separated by d_{LR} with width σ :

$$I_4 \propto \exp\left(-\frac{d_{\text{LR}}^2}{2\sigma^2}\right) \quad (16)$$

The overlap is exponentially suppressed by the separation.

Proof. For $w_L \propto \exp(-(\chi - \chi_L)^2/(2\sigma^2))$ and similar for w_R centered at χ_R :

$$w_L^2 w_R^2 \propto \exp\left(-\frac{(\chi - \chi_L)^2 + (\chi - \chi_R)^2}{\sigma^2}\right) \quad (17)$$

$$= \exp\left(-\frac{2\chi^2 - 2\chi(\chi_L + \chi_R) + \chi_L^2 + \chi_R^2}{\sigma^2}\right) \quad (18)$$

Integrating over χ gives exponential suppression $\propto \exp(-d_{\text{LR}}^2/(2\sigma^2))$. $\square$

4.2 Gate 1 Constraint

Proposition 4.3 (Required Suppression). [Dc] From OPR-21, Gate 1 requires:

$$I_4 \approx 2 \times 10^{-3} \text{ GeV} \quad (19)$$

to match the target $X_{\text{EDC}} \approx X_{\text{target}} = G_F m_e^2$.

Corollary 4.4 (Required Separation). [Dc] With $\sigma \sim \delta \approx 0.5 \text{ GeV}^{-1}$, achieving $I_4 \sim 10^{-3} \text{ GeV}$ requires:

$$\frac{d_{\text{LR}}^2}{2\sigma^2} \sim \ln(10^3) \approx 7 \quad \Rightarrow \quad \frac{d_{\text{LR}}}{\delta} \sim \sqrt{14} \approx 4 \quad (20)$$

Including polynomial prefactors and the actual mode shapes from BVP:

$$\boxed{\frac{d_{\text{LR}}}{\delta} \approx 5\text{--}10} \quad (21)$$

Key insight: The G_F constraint imposes $d_{\text{LR}}/\delta \sim O(10)$. The BVP scan found $d_{\text{LR}}/\delta = 8$ passes all gates. This is **not arbitrary** — it is the value that produces the required I_4 suppression.

4.3 Why 8δ Specifically?

The sensitivity analysis (OPR-21c) showed:

- LR elasticity = -6.5 (dominant control)
- 10% change in $d_{\text{LR}} \rightarrow 65\%$ change in X_{ratio}

The window $d_{\text{LR}}/\delta \in [7, 9]$ passes Gate 1. The center of this window is ≈ 8 . This is:

$$d_{\text{LR}} = 8\delta = 8 \times 0.533 \text{ GeV}^{-1} = 4.26 \text{ GeV}^{-1} = 0.84 \text{ fm} \quad (22)$$

5 The r_p Coincidence

5.1 Numerical Match

Quantity	Value	Source
d_{LR} (from BVP scan)	0.84 fm	[Cal] (Gate 1)
r_p (proton charge radius)	0.841 fm	[BL] (CODATA)
Match	0.1%	—

5.2 Possible Interpretations

1. **Coincidence:** Both are $O(\text{QCD scale})$, so agreement within factors of 2 is not surprising. Many QCD lengths are $\sim 1 \text{ fm}$.
2. **Common origin:** If the proton's structure (Y-junction) also determines the chiral localization geometry, both would be set by the same length scale.
3. **Constraint propagation:** Perhaps r_p is *defined* by this chiral scale, not vice versa. The proton radius emerges from the same 5D geometry that sets d_{LR} .

Status of r_p Coincidence

The match $d_{\text{LR}} \approx r_p$ is:

- **NOT derived** — we don't have a mechanism connecting them
- **NOT accidental** — the O(10%) precision is suggestive
- **Hypothesis-forming** — suggests Y-junction geometry

To upgrade: Derive r_p and d_{LR} from the same 5D junction geometry. Show that the localization positions depend on the junction arm length, which equals r_p .

6 Derivation Summary

Result	Status	Comment
Single domain wall $\rightarrow d_{\text{LR}} = 0$	[Der]	Standard result
$d_{\text{LR}} > 0$ requires additional structure	[Der]	Two walls, Yukawa, or junction
Gate 1 imposes $d_{\text{LR}}/\delta \sim 5\text{--}10$	[Dc]	From $I_4 \sim 10^{-3}$ GeV
BVP scan gives $d_{\text{LR}}/\delta = 8$	[Cal]	Fit to target X_{EDC}
$d_{\text{LR}} = 0.84 \text{ fm} \approx r_p$	[I]	Coincidence (suggestive)
Junction geometry $\rightarrow d_{\text{LR}} \sim r_p$	[P]	Hypothesis

7 Stoplight Verdict

Epistemic Status: **YELLOW** [Dc/Cal]

What is derived [Der]:

- Single symmetric domain wall gives $d_{\text{LR}} = 0$
- Non-zero separation requires two walls, Yukawa, or junction geometry
- Overlap integral I_4 is exponentially sensitive to d_{LR}

What is derived conditional [Dc]:

- Gate 1 constraint imposes $d_{\text{LR}}/\delta \sim 5\text{--}10$
- Two-wall or junction mechanism can produce this separation

What is calibrated [Cal]:

- Specific value $d_{\text{LR}}/\delta = 8$ from parameter scan

What is a hypothesis [P]:

- Y-junction geometry produces $d_{\text{LR}} \sim r_p$
- Proton charge radius equals chiral separation scale

Upgrade path YELLOW $\rightarrow$ GREEN:

1. Derive the two-wall separation (or junction geometry) from 5D action
2. Show that $d_{\text{LR}} = r_p$ follows from proton = Y-junction
3. Compute r_p itself from the junction geometry

8 Conclusion

We have established:

1. **A single domain wall cannot produce L-R separation.** Both chiralities (when present) localize at the wall center.
2. **To get $d_{\text{LR}} > 0$, we need:**
 - Two domain walls at different positions, OR
 - Yukawa coupling to a non-trivial Higgs profile, OR
 - Y-junction geometry (EDC-specific)
3. **The G_F gate imposes $d_{\text{LR}}/\delta \sim 5\text{--}10$.** This is a constraint from requiring $I_4 \sim 10^{-3}$ GeV. The BVP scan found $d_{\text{LR}}/\delta = 8$ satisfies all gates.
4. **The coincidence $d_{\text{LR}} \approx r_p$ is suggestive but not derived.** It motivates the Y-junction interpretation but does not prove it.

Defendable claim: The value $d_{\text{LR}}/\delta \approx 8$ is **not arbitrary tuning** — it is constrained by the G_F gate to lie in the range $[5, 10]$. The specific value 8 is the center of this window. The coincidence with r_p suggests deeper structure but remains a hypothesis.